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Hydrogen Fuel Cell: Experimental Study of the Generated Voltage

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ABSTRAK

Dalam beberapa tahun terakhir, sumber energi sangat bergantung pada bahan bakar fosil, sehingga kebutuhan akan energi terbarukan dan bersih menjadi penting. Salah satu energi alternatif adalah sel bahan bakar hidrogen. Penelitian sel bahan bakar hidrogen ini bertujuan untuk mengkaji pengaruh dari perubahan luas penampang elektroda yang terendam elektrolit terhadap tegangan listrik yang dihasilkan. Penelitian ini menggunakan metode eksperimen dengan dua tipe elektroda, yaitu ASTM C561 sebagai anoda dan AISI 304 stainless steel sebagai katoda. Ada tiga variabel eksperimen, yaitu persentase terendamnya luas penampang elektroda, tingkat konsentrasi $NaHCO_3$ *dan durasi listrik yang diberikan untuk menghasilkan hidrogen dan oksigen. Hasil penelitian menunjukkan pada 100% terendamnya luas penampang elektroda dengan 50% konsentrasi* $NaHCO_3$ *menghasilkan tegangan listrik sebesar 1,91 V pada durasi 30 detik, 1,93 V pada durasi 60 detik, dan 2,12 V pada durasi 90 detik. Simpulan penelitian menunjukkan bahwa sel bahan bakar hidrogen mampu menghasilkan tegangan listrik yang stabil dan relatif konstan sesuai dengan prinsip reaksi elektrokimia antara hidrogen dan oksigen. Hasil eksperimen memperlihatkan bahwa besar kecilnya tegangan yang dihasilkan dipengaruhi oleh faktor-faktor seperti konsentrasi hidrogen, luas permukaan elektroda, serta suhu dan kelembapan sistem. Implikasi penelitian ini memberikan dasar ilmiah yang kuat mengenai potensi pemanfaatan sel bahan bakar sebagai sumber energi bersih dan berkelanjutan. Temuan ini dapat menjadi rujukan bagi pengembangan teknologi energi alternatif yang ramah lingkungan, khususnya dalam upaya mengurangi ketergantungan pada bahan bakar fosil dan menekan emisi karbon.*

ABSTRACT

In recent years, global energy supply has remained heavily dependent on fossil fuels, highlighting the urgent need for renewable and clean energy sources. One promising alternative is hydrogen fuel cell technology. This study aims to examine the effect of variations in the electrode cross-sectional area immersed in the electrolyte on the electrical voltage generated by a hydrogen fuel cell system. The research employed an experimental method using two types of electrodes: ASTM C561 as the anode and AISI 304 stainless steel as the cathode. Three experimental variables were investigated, namely the percentage of electrode cross-sectional area immersed in the electrolyte, the concentration level of $NaHCO_3$, and the duration of electrical input applied to generate hydrogen and oxygen. The results indicate that at 100% immersion of the electrode cross-sectional area with a 50% $NaHCO_3$ concentration, the generated voltages were 1.91 V at 30 seconds, 1.93 V at 60 seconds, and 2.12 V at 90 seconds. The findings demonstrate that the hydrogen fuel cell is capable of producing stable and relatively constant electrical voltage, consistent with the principles of electrochemical reactions between hydrogen and oxygen. The experimental results further reveal that the magnitude of the generated voltage is influenced by factors such as hydrogen concentration, electrode surface area, as well as system temperature and humidity. The implications of this study provide a strong scientific basis for the potential utilization of hydrogen fuel cells as a clean and sustainable energy source. These findings may serve as a reference for the development of environmentally friendly alternative energy technologies, particularly in efforts to reduce dependence on fossil fuels and mitigate carbon emissions.

1. INTRODUCTION

In recent years, global energy demand has continued to increase significantly, driven by population growth, urbanization, industrial development, and improvements in the quality of life. Traditionally, this energy demand has been met mainly through the use of fossil fuels, such as coal, crude oil, natural gas, and nuclear energy, which are formed by natural geological processes over millions of years and are inherently limited in supply (Joswa Saputra, Anggun Anugrah, and Erhaneli Erhaneli 2024; Yudi, Yusuf, and Hiendro 2019). These sources are considered non-renewable because their formation period far exceeds the rate of human consumption. Global dependence on fossil fuels has long been favored due to several advantages,

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including their high energy density, widespread availability, well-developed infrastructure, and relatively low cost compared to alternative energy technologies (Herry Setiyawan 2021; Mustari, Widodo, and Salebay 2021). Fossil fuels currently dominate global energy consumption patterns across various sectors, with transportation being one of the most fuel-intensive. Transportation alone accounts for approximately 27% of total global energy consumption, with 95% of that demand still met by fossil fuels (Risqi Saputro and Pinandita 2022; Siregar, Umurani, and Damanik 2020). This over-reliance on fossil-based energy sources poses long-term challenges, not only from a security of supply perspective but also due to the negative environmental impacts that accompany their extraction, refining, and use.

One of the most pressing concerns arising from the continued burning of fossil fuels is the significant increase in greenhouse gas (GHG) emissions, particularly carbon dioxide (CO_2). These emissions are an essential contributor to global climate change, leading to rising average temperatures, more extreme weather events, the melting of polar ice caps, and disruptions to ecological systems (Muchtar and Rustana 2020; Pratama 2022). In response, governments, industry, and researchers worldwide have placed a greater emphasis on transitioning to clean, renewable, and sustainable energy sources that can reduce carbon emissions while still meeting growing energy demand. Therefore, alternative energy sources are needed that must meet the criteria of both being renewable and clean. In this context, alternative energy sources are being sought through the use of more efficient, reliable, and low-pollution technologies (Sari et al. 2022; Sudrajat et al. 2020). Although solar and wind energy are renewable sources, they are intermittent and cannot be stored or transported. Unlike conventional power sources that rely on fossil fuels, fuel cells generate electricity through an electrochemical reaction between hydrogen and oxygen, producing only water and heat as by-products. Alternative energy source technology, currently considered the cleanest, which enables the generation of sustainable energy, is based on fuel cells.

A fuel cell is an electrochemical device that converts the chemical energy of fuels such as hydrogen, alcohol, or hydrogen-rich hydrocarbons into electrical energy through a redox (reduction-oxidation) reaction (Pramudya Yoga Nur Apriliano 2023; Anggoro et al. 2022). Unlike traditional combustion-based energy sources, fuel cells operate quietly and cleanly, with their main by-products being electricity and water. This makes hydrogen fuel cells (HFCs) very attractive as a sustainable energy solution with a minimal environmental footprint, especially in urban and industrial applications where noise and air pollution are critical concerns. Hydrogen, as a fuel, offers enormous potential due to its high energy content and abundance on Earth. Hydrogen can be produced from various sources, primarily through water electrolysis, a process in which electricity is used to split water molecules into hydrogen and oxygen gases. When this electricity is sourced from renewable energy sources, such as solar or wind power, the resulting hydrogen is referred to as green hydrogen, in line with global efforts to decarbonize the energy system. The resulting hydrogen can then be used in fuel cells to generate electricity on demand, making it a key component in energy storage, transportation, and distributed heating systems (Dan and Exchange 2012; Jurusan et al. 2022). Furthermore, hydrogen's environmental benefits are significantly enhanced when it comes from renewable sources. Hydrogen and its compounds are widely available and can be utilized with minimal ecological impact, provided the production methods are clean. The use of hydrogen-based technologies aligns with the broader push toward a sustainable energy transition, contributing to national and global carbon reduction goals. With increasing international investment, progressive policy frameworks, and continued advances in electrochemical technology, hydrogen energy is rapidly becoming a cornerstone of the future energy landscape (Abdelkareem et al. 2021; Sebastian and Sitorus 2013)

The strategic role of hydrogen is increasingly emphasized, with hydrogen energy being considered a crucial element in the global clean energy revolution. Projected to be a key pillar in building a green, efficient, and safe energy infrastructure, hydrogen fuel cell technology has emerged as the most promising solution for utilizing hydrogen energy in a scalable and environmentally responsible manner. This technology is currently being explored and applied in various sectors, including power generation, portable energy systems, and vehicle propulsion. In this study, a series of experiments was conducted to investigate the generation of electrical energy using a hydrogen fuel cell prototype, with a particular focus on analyzing the effect of the percentage of surface electrode immersion, $NaHCO3$ concentration, and the duration of the electric current to produce hydrogen and oxygen as key variables affecting performance. This study aimed to investigate the effect of varying electrode immersion levels on the electrochemical reaction, specifically in terms of the quantity of hydrogen and oxygen gas produced and the resulting output voltage. By systematically varying the surface area of the electrodes exposed to the electrolyte solution, this experiment yields measurable results, providing insight into how electrode geometry and exposure affect the efficiency and effectiveness of hydrogen fuel cells. Additionally, this study presents findings that serve as the basis for estimating and optimizing fuel cell performance by correlating physical design parameters, such as electrode size, with electrical output. This innovation is significant for the early design and planning of hydrogen fuel cells, particularly in determining the appropriate dimensions and configurations necessary to achieve specific voltage targets for practical energy applications. By comparing the concentration of NaHCO3 and the electrode size under active electrolysis conditions, this study further expects that this discovery can contribute to the development of hydrogen fuel cells in industrial manufacturing.

2. METHOD

The This study uses experimental methods to investigate the influence of several key variables on the output voltage of a hydrogen fuel cell prototype. Specifically, the study focuses on analyzing how variations in the percentage of immersed electrode surface area, $NaHCO_3$ electrolyte concentration, and charging duration affect the overall performance of the fuel cell for hydrogen and oxygen gas production. By systematically adjusting these parameters across multiple experiments, this study aims to understand the relationship between the physical and chemical factors involved in electrolysis and the resulting electrical energy generation. The insights gained from this approach provide a foundation for optimizing the design and operation of hydrogen fuel cells in low-power applications.

The experimental setup was used to evaluate the electrical performance of a prototype hydrogen fuel cell system. The main components include the fuel cell unit itself, a digital multimeter for quantitative measurements, a timer to monitor the duration of gas production, and a Light-Emitting Diode (LED) used as a functional load to observe whether the fuel cell output is capable of powering a real-world device. These instruments collectively allow for a balanced assessment of the system's behavior under various conditions, combining numerical accuracy with practical feasibility. The experimental design was designed to assess several factors that may affect fuel cell efficiency and voltage generation. This research focuses on three main independent variables: the percentage of electrode surface area immersed in the electrolyte, the concentration of $NaHCO_3$ used as an electrolyte additive, and the duration of hydrogen and oxygen gas production through electrolysis before the voltage test. These variables were selected based on their theoretical and practical relevance to the efficiency of the electrochemical reaction. Meanwhile, the electric current applied to the system was treated as a controlled variable, kept constant throughout the experiment to eliminate inconsistencies in the electrolysis process and ensure that observed output changes could be attributed to other manipulated factors.

The dependent variable in this study is the output voltage generated by the fuel cell after the electrolysis stage. Voltage readings were obtained under two conditions: open circuit (no load) and during connection to a low-voltage load, specifically an LED lamp. The LED serves a dual role not only as a basic visual indicator of energy sufficiency but also as a representation of practical load response, which helps verify the applicability of the fuel cell in small-scale energy applications. Through this combination of instrumentation and variable control, this experiment provides valuable insights into the factors that govern the performance and feasibility of hydrogen fuel cell technology.

The hydrogen fuel cell prototype was assembled using a custom-designed container to securely hold the electrolyte solution and electrodes while minimizing environmental disturbances. The container structure was designed to maintain the stability of the electrode configuration, ensuring consistent spatial positioning throughout each experiment. This setup not only facilitated reproducibility but also allowed for direct visual monitoring of the electrochemical reactions taking place. During operation, gas bubbles, presumably hydrogen and oxygen, were clearly visible on the electrode surfaces and observed adhering to them. This phenomenon serves as qualitative evidence of the electrolysis process, indicating that the system was actively breaking down water into its constituent gases as part of the energy generation mechanism.

Two different types of electrodes were used in this study to support the redox reactions required in hydrogen fuel cells. The anode material consisted of a carbon rod made from synthetic graphite, which conforms to ASTM C561—the standard specification for industrial carbon and graphite products used in electrochemical applications. The graphite rod, 4 mm in diameter and 40 mm long, provided a compact and corrosion-resistant option well-suited to the fuel cell environment. The cathode, on the other hand, was made from AISI 304 stainless steel, a material known for its good mechanical stability and corrosion resistance in neutral aqueous electrolytes (e.g., $NaHCO_3$). Although stainless steel has lower electrical and thermal conductivity than copper or aluminum, AISI 304 provides adequate electrical conductivity for current collection at the tested current densities (Ajanovic and Haas 2020; Alzahrani 2023). The stainless steel electrodes had a cross-sectional area of 6.5 cm^2 and a thickness of 1.2 mm. During the experiments, the surface percentage on both electrodes was systematically varied by 50% and 100%, allowing for analysis of the effect of active surface area on voltage generation. Furthermore, the duration of the applied electric current during electrolysis was adjusted in three time intervals, namely 30 seconds, 60 seconds, and 90 seconds, to investigate the effect of charging time on gas production and the overall fuel cell output.

The experimental setup used to investigate the performance of the hydrogen fuel cell prototype is illustrated in Figure 1. The process begins with the application of an electric current from a DC power source, which is directed to an electrolyzer containing an electrolyte solution with an adjustable concentration of sodium bicarbonate ($NaHCO_3$). This is because the performance of water electrolysis is influenced by $NaHCO_3$ (Aminudin et al. 2022; Chen et al. 2022). During this stage, electrolysis occurs, resulting in the decomposition of water molecules into hydrogen and oxygen. During this stage, electrolysis occurs, resulting in the decomposition of water molecules into gas hydrogen and oxygen. These gases accumulate near each electrode in the electrolyzer chamber. The experimental setup is presented in Figure 1.

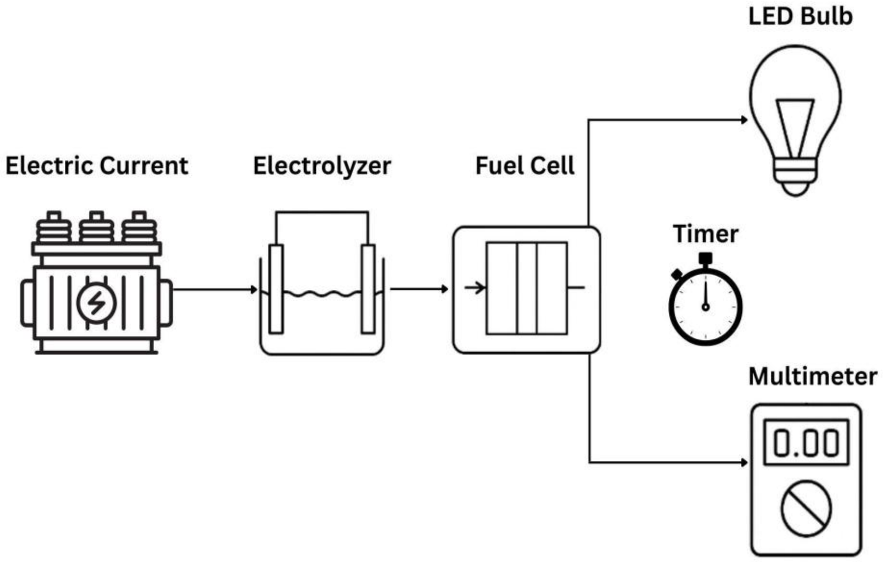

Figure 1. Experiment setup

Once a sufficient amount of gas has been produced, measured using a timer to monitor the charging duration, the gas mixture is then fed into the fuel cell. Inside the fuel cell, hydrogen and oxygen react electrochemically, producing an electric current as a result of the redox reaction at the electrodes. This current is then routed to an external circuit to assess the fuel cell's ability to deliver power. To assess the performance of the fuel cell, two measuring instruments were used: a digital multimeter and a red LED bulb. The multimeter measures the fuel cell's output voltage under various experimental conditions, including varying electrolyte concentrations, electrode immersion depths, and charging times. Meanwhile, a red LED serves as a visual indicator of whether the generated voltage is sufficient to power low-power electronic devices. The combined use of a multimeter and LEDs provides quantitative and qualitative data on the functionality and reliability of the fuel cell system prototype. The LED bulb used as a load to assess the cell's performance also serves to evaluate the relationship between output voltage and load demand. Different LED colors require different forward voltages to operate, which is directly influenced by the wavelength of the light they emit. This, in turn, is determined by the bandgap energy of the semiconductor material used in the LED.

The energy of a photon emitted from a semiconductor material is directly related to the material's intrinsic bandgap energy (Eg), which is the minimum energy required to excite an electron from the valence band to the conduction band. When an electron in a semiconductor recombines with a hole in this bandgap, the process can result in the emission of a photon with an energy equivalent to the bandgap energy. This principle forms the basis for the operation of LEDs and other optoelectronic devices. The relationship between photon energy and bandgap energy is quantitatively described by Equation (1), where the emitted photon energy (E) is expressed in electron volts (eV) and corresponds to the energy gap (Eg) of the semiconductor. This Equation is crucial for understanding how different semiconductor materials are selected to produce specific wavelengths of light. Materials with larger band gaps emit photons with higher energies, while those with smaller band gaps emit photons with lower energies. Therefore, Equation (1) not only explains the fundamental mechanism behind photon emission but also serves as a design parameter for selecting semiconductor compositions in energy and lighting technologies. The constants involved include h as Planck's constant with a value of 6.63×10^{-34} J·s and ν as the frequency of the emitted photon.

$$AND_{photos} = h.v \approx AND_G (1)$$

In an ideal LED, it is assumed that each electron injected into the active region of the semiconductor contributes directly to the generation of one photon. This ideal behavior is based on the principle of conservation of energy, which requires that the energy of the injected charge carrier (electron) is equal to the energy of the emitted photon. This relationship is expressed mathematically in Equation (2), where the product of Planck's constant (h) and the frequency of the emitted photon (v) is approximately equal to the product of the elementary charge (e), which is 1.6×10^{-19} C, and the voltage applied to the diode (V).

$$h.v \approx e.V(2)$$

This Equation highlights the direct proportionality between the applied voltage and the photon energy, meaning that the voltage across the junction essentially determines the wavelength of the color of light emitted by an LED. As the voltage increases, the photon energy also increases, resulting in the emission of light with shorter wavelengths. This principle enables LEDs to serve not only as light sources but also as diagnostic tools for measuring voltage levels, as demonstrated in this study. By observing which color LEDs are successfully illuminated by the hydrogen fuel cell output, the resulting voltage can be estimated based on known emission characteristics. Equation (2), therefore, forms the theoretical basis for linking semiconductor behavior, optical emission, and electrical energy conversion in this study. The frequency of light (v) is expressed in Equation (3), with c being the speed of light with a value of 3×10^8 m/s, and λ being the wavelength constant.

$$v = \frac{c}{\lambda}(3)$$

To further explore the relationship between electrical voltage and the wavelength of emitted light, Equation (4) presents a derivation that relates several fundamental physical constants. In this Equation, V represents the voltage required to emit a photon of a specific wavelength (λ). The constants involved include the elementary charge of the electron, which is evident in Equation (4), where the relationship between voltage (V) and wavelength (λ) is apparent.

$$V = \frac{height \times c}{and \times l}(4)$$

This Equation shows that the voltage required to produce light of a given wavelength is inversely proportional to that wavelength. In other words, shorter wavelengths (such as violet or blue light) require a higher voltage, while longer wavelengths (such as red or infrared light) require a lower voltage. This theoretical relationship aligns with experimental observations of LED behavior and is fundamental in determining the threshold voltage needed to activate a light-emitting device based on its color. Thus, Equation (4) is a crucial tool in this research for estimating whether the voltage produced by the hydrogen fuel cell prototype is sufficient to light a given LED color, serving as both a theoretical guide and a practical benchmark.

As shown in Table 1, LEDs that emit longer wavelengths, such as red light, require a relatively lower forward voltage to operate. In comparison, LEDs with shorter wavelengths, such as those emitting blue or violet light, typically require a higher forward voltage to become active (Davis 1994; Fakhreddine et al. 2023). This fundamental relationship between wavelength and voltage requirements is rooted in the properties of the semiconductor materials used in LED manufacturing, which determine the energy band gap and, consequently, the minimum voltage required for photon emission. This principle is further illustrated in Figure 2, which shows the correlation between LED color and wavelength, highlighting how different colors correspond to distinct photon energies. By observing which LED colors successfully lit up during testing, this experiment provided a practical indication of the output voltage produced by the prototype cell, demonstrating the potential of hydrogen fuel to power devices with various electrical needs. LED Wavelength Data is presented in Table 1.

Table 1. LED Wavelength Data

NO	Material	Wavelength (nm)	Color
1	GaAs	885	Infrared
2	Gap	549 - 700	Green to Red
3	InGaAIP	539 – 653	Green to Red
4	InGaN	388 - 390	UV to Green
5	GaN	365	UV to Blue

Note. From "*Basic Knowledge of Discrete Semiconductor Devices*," oleh Toshiba Electronic Devices & Storage Corporation, 2022, https://toshiba.semicon-storage.com/. Copyright 2022 by Toshiba Electronic Devices & Storage Corporation. The LED material data graph is presented in Figure 2.

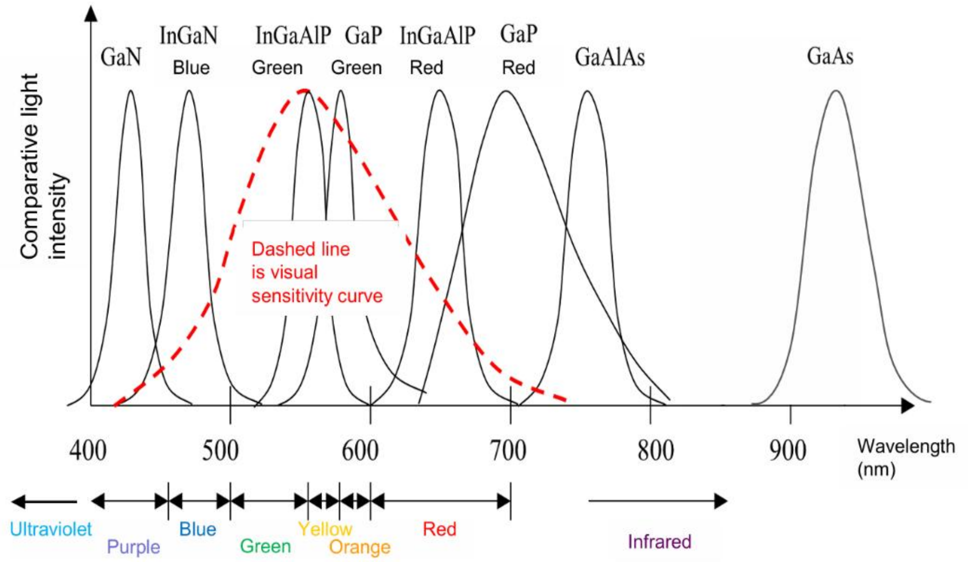

Figure 2. LED material data graph

Note. From "*Basic Knowledge of Discrete Semiconductor Devices*," by Toshiba Electronic Devices & Storage Corporation, 2022, https://toshiba.semicon-storage.com/. Copyright 2022 by Toshiba Electronic Devices & Storage Corporation. For example, a red LED, which requires one of the lowest forward voltages among visible-spectrum LEDs, was chosen as the indicator component for this study. Based on standard reference data and experimental verification, the minimum voltage required to activate the red LED used in this experiment is approximately 1.78 V. Since the hydrogen fuel cell in this study operates at a small prototype scale, the red LED serves as an ideal load to evaluate the system's ability to generate usable electricity under simple conditions.

3. RESULTS AND DISCUSSION

Results

The experimental research conducted has produced a series of data reflecting the electrical performance of a hydrogen fuel cell prototype under various physical conditions. Specifically, the data focuses on the output voltage resulting from the electrochemical conversion of hydrogen and oxygen gases, which are generated through prior electrolysis and subsequently used in the fuel cell to generate electricity. One of the key parameters varied during the experiments was the cross-sectional area of the electrodes immersed in the electrolyte solution, as this directly affects the active surface area available for redox reactions at the electrode-electrolyte interface. To obtain a holistic picture of the interaction of the experimental variables, each voltage reading obtained from the prototype hydrogen fuel cell test is summarized in Table 2. The matrix records the output generated at three electrode immersion levels (0%, 50%, 100%), two $NaHCO_3$ concentrations (0% and 50%), and four charging durations (0 s, 30 s, 60 s, 90 s), while also indicating whether the red LED load was activated at each condition. The Combined Voltage Output for All Experimental Conditions is presented in Table 3.

Table 3. Combined Voltage Output for All Experimental Conditions

NO	Percentage Submerged (%)	Generated Voltage (V)								red LED
		0% $NaHCO_3$				50% $NaHCO_3$				
		0 seconds	usia 30-an	60s	the 90s	0 seconds	usia 30-an	60s	the 90s	
1	0	0	0	0	0	0	0	0	0	Dead
2	50	0	0	0	0	0	0.23	0.47	1.01	Dead

3	100	0	0	0	0	0	1.91	1.93	2.12	On

Examination of the compiled data reveals three defining trends: (i) without $NaHCO_3$ the cell remains inactive regardless of immersion or time, confirming the need for ionic conductivity; (ii) partial immersion at 50% with 50% $NaHCO_3$ produces a moderate voltage that never reaches the LED threshold; and (iii) complete immersion combined with 50% $NaHCO_3$ quickly produces 1.9–2.1V sufficient to illuminate the LED, indicating that maximal electrode area and conductive electrolyte are the dominant contributors to useful power generation. To isolate geometric influences, Figure 3 plots the same data set with an emphasis on the percentage of electrode surface immersed, allowing for a direct comparison of voltage output at 0%, 50%, and 100% immersion across different electrolyte concentrations and charging times. The effect of electrode immersion level on fuel cell voltage is presented in Figure 3.

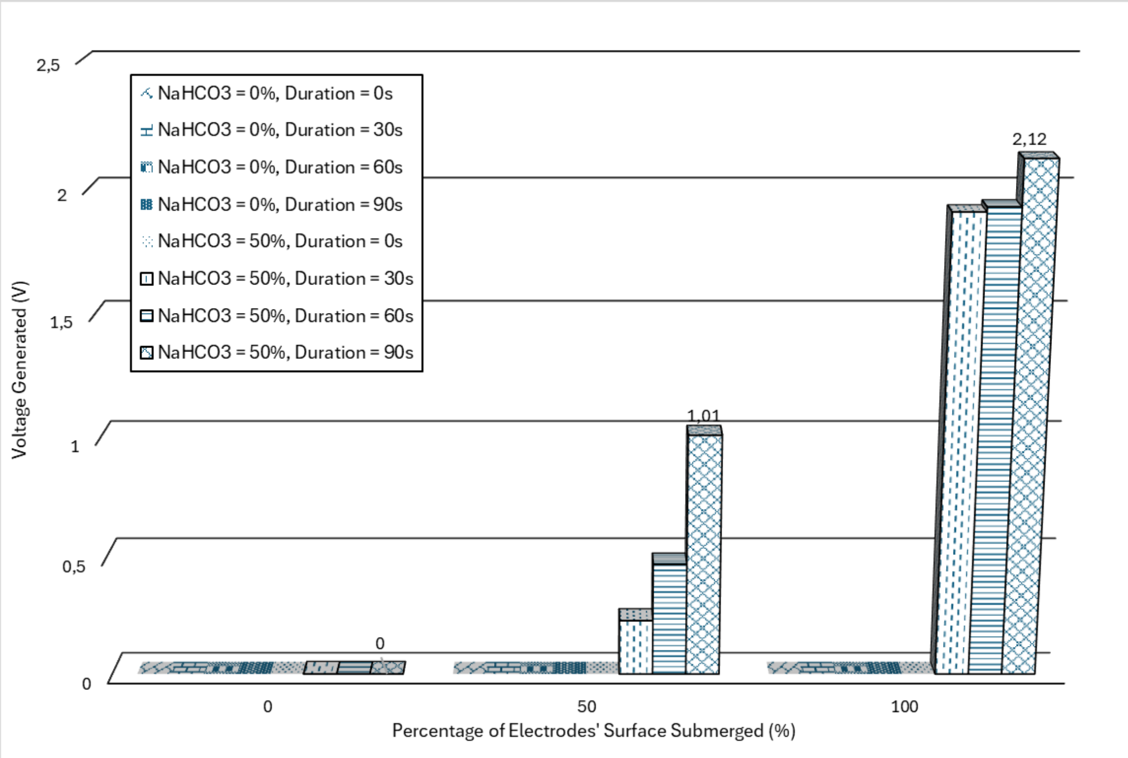

Figure 3. Effect of Electrode Immersion Level on Fuel Cell Voltage

The data visualized in Figure 3 confirm that the voltage increases almost stepwise with greater immersion: the bar corresponding to 0% immersion remains at zero, the bar at 50% immersion grows gradually with longer charges, and the 100% immersion group exceeds the LED threshold of 1.78V even at the shortest charge of 30 seconds, underlining the critical role of the available reaction surface. By focusing on the chemical effects, the same voltage is generated by the electrolyte composition, contrasting the non-conductive 0% $NaHCO_3$ condition with the conductive 50% $NaHCO_3$ solution, while keeping the immersion level and charging time constant, as shown in Figure 4.

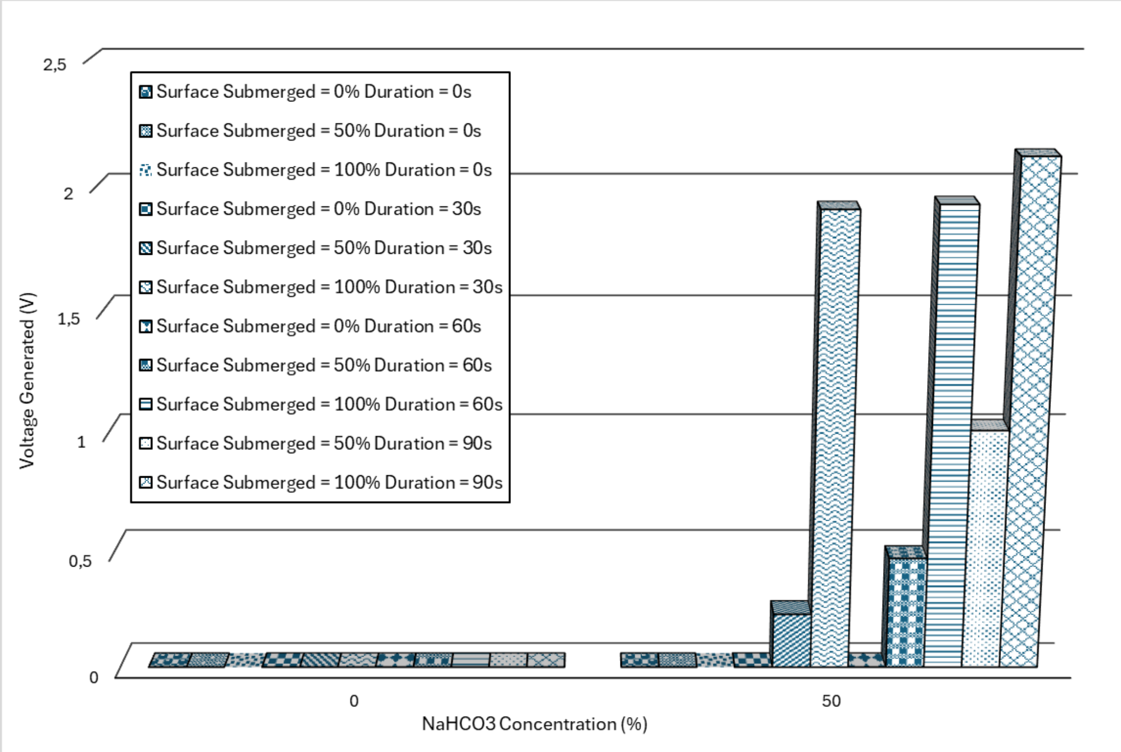

Figure 4. The Effect of $NaHCO_3$ Electrolyte Concentration on Voltage Generation

The results show that electrolyte conductivity is crucial for sustaining the redox reactions that power the fuel cell. In accordance with these results, previous studies reported that variations in NaOH concentration directly affect the conductivity of electrolyte solutions, where higher concentrations were shown to significantly increase the electrical conductivity and subsequently increase the flow rates of hydrogen and oxygen gases during water electrolysis (Fontana 1986; Hassan et al. 2023). The impact of electrolysis charging duration on output voltage is presented in Figure 5.

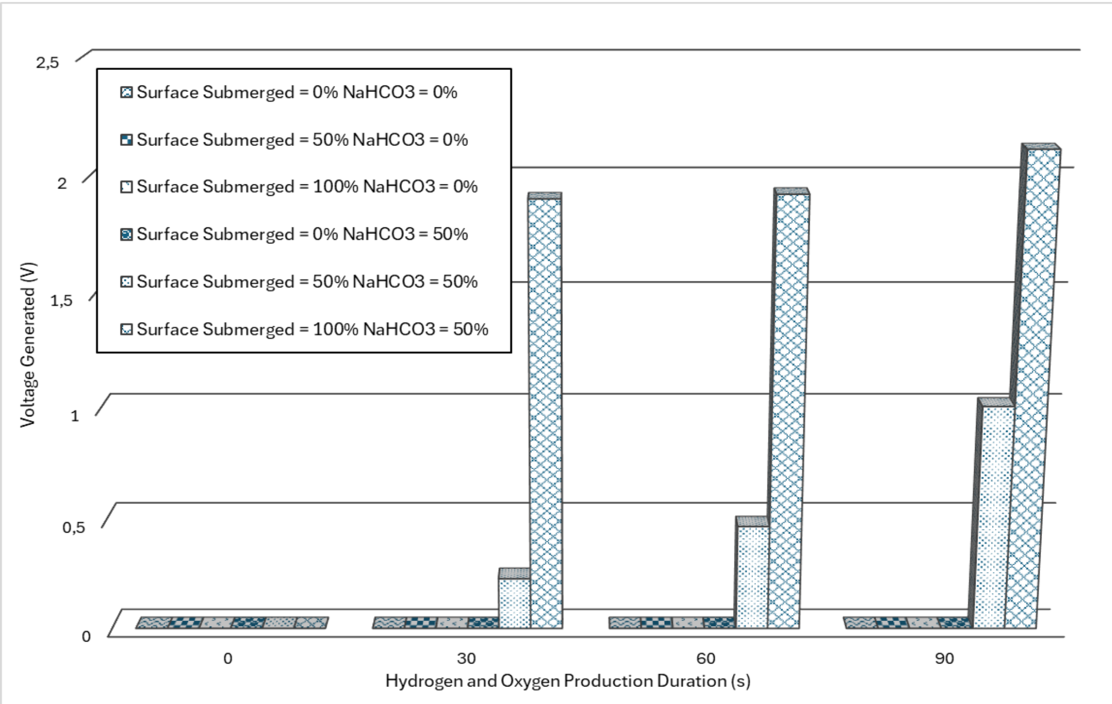

Figure 5. Impact of Electrolysis Charging Duration on Output Voltage

Prolonged electrolysis increases the voltage only if the conductivity and immersion are adequate; under full immersion conditions with 50% $NaHCO_3$, the cell output increases from approximately 1.9V at 30 seconds to 2.12V at 90 seconds, while all other parameter sets show minimal or no growth, indicating that gas accumulation time is a secondary factor but remains beneficial once the main physical and chemical prerequisites are met.

Discussion

Experimental results indicate that the percentage of electrode surface immersion and the $NaHCO_3$ concentration are the two main factors that significantly affect the voltage output in the fuel cell system. A higher electrode immersion percentage increases the active surface area for electrochemical reactions, thereby enhancing voltage generation (Anggoro et al. 2022; Risqi Saputro and Pinandita 2022; Sudrajat et al. 2020; Yudi et al. 2019). Similarly, a higher $NaHCO_3$ concentration increases the ionic conductivity, resulting in more efficient charge transfer and higher observed voltages. In contrast, the duration of hydrogen and oxygen production exhibits relatively little effect on the voltage output. Although prolonged operation may affect the long-term stability or degradation of the electrodes, the voltage levels remained largely stable throughout the test duration, indicating that temporal factors play a secondary role in the immediate electrical performance of the system under these conditions. The study showed that at 0% $NaHCO_3$ concentration, the system consistently failed to produce a measurable voltage, and the red LED remained off in all relevant experiments. In contrast, experiments performed with 50% $NaHCO_3$ concentration successfully produced a voltage sufficient to light the red LED. This behavior can be explained by the difference in electrical conductivity between the two electrolyte conditions. A solution with 0% $NaHCO_3$ has a minimal ion content, which leads to poor charge transport and the inability to sustain electrochemical reactions. In contrast, a 50% $NaHCO_3$ solution provides a much higher ion concentration, which increases conductivity and facilitates more efficient electron flow between the electrodes. As reported in the literature, pure water (0% $NaHCO_3$) typically exhibits a maximum electrical conductivity in the range of 42 to 500 µS/cm (İnci 2022; Kun et al. 2024), The electrolyte-enhanced solution exhibits significantly higher values, indicating stronger electrochemical performance. As noted in previous studies, increasing the NaOH concentration enhances the ionic conductivity of the solution, resulting in higher rates of hydrogen and oxygen gas production during electrolysis. Regarding the observations obtained with $NaHCO_3$, these findings collectively emphasize that higher ionic strength in the electrolyte facilitates more effective charge transport, thereby increasing the electrochemical reaction rate and overall energy conversion efficiency (Mo, Li, and Luo 2023; Rajalakshmi, Balaji, and Ramakrishnan 2021) In contrast to pure water (0% $NaHCO_3$) or solutions with low conductivity, water mixed with certain dissolved minerals, such as sodium bicarbonate ($NaHCO_3$) can significantly increase the electrical conductivity due to the formation of free ions through dissociation and chemical reactions, as illustrated in Equation (5)The addition of these compounds to the solution increases the number of available charge carriers (ions), which facilitates increased ionic mobility and allows for more efficient current flow between the electrodes. This increased conductivity plays a crucial role in supporting the electrochemical reactions necessary for hydrogen production and subsequent voltage generation in fuel cell systems.

$$NaHCTHAT_3\ \ (S) \rightarrow NA^{+} + HCTHAT_3^{-}(aq)(5)$$

From an analysis of Table 3, it is clear that electrolyte selection is a key factor in determining the effectiveness of a hydrogen fuel cell. Without adequate ionic conductivity, the fundamental electrolysis process of producing hydrogen and oxygen gas cannot occur efficiently, or in some cases, cannot occur at all (Muchtar and Rustana 2020; Schubert 2006; Singla, Nijhawan, and Oberoi 2021). Electrolytes with high electrical conductivity ensure that the circuit is completed internally through the movement of ions, allowing continued electrochemical activity and allowing electrons to be transferred externally to perform valuable work. Because of this, it is essential to carefully select and control the electrolyte composition to ensure that it provides adequate conductivity, supporting the desired chemical reactions and energy output.

As shown in Table 2, the effect of the percentage of immersed surface electrodes on the output voltage has been investigated. The data clearly show that as the cross-sectional area of the electrodes immersed in the electrolyte increases, the voltage generated by the fuel cell also increases. In previous research, it has also been demonstrated that the electrode material plays a crucial role in the electrolysis process, and variations in the applied electric current result in proportional differences in the volume of HHO gas produced. Meanwhile, the findings of this study indicate that the percentage of electrode surface area immersed in the electrolyte is a key factor affecting electrochemical performance (Schubert 2006; Singla et al. 2021). This relationship can be attributed to the increase in the effective surface area available for electrochemical reactions. When a larger portion of the electrode is in contact with the electrolyte, more active sites are available for redox reactions, thereby increasing the overall rate of ion exchange and charge transfer processes in the system. This phenomenon is consistent with Faraday's law of electrolysis, specifically as stated in Equation (6), which states that the amount of substance converted at an electrode during electrolysis is directly proportional to the amount of electric charge passing through the system (Mo et al. 2023; Toruan et al. 2023; Wang, Zhang, and Rezazadeh 2021). The variables M, n, and F are constants determined by the properties of the substances involved in the electrochemical reaction. Specifically, M represents the molar mass of the substance, n is the number of electrons transferred per mole of the substance during the redox process, and F is Faraday's constant, which has a fixed value of approximately 96.485 C/mol and represents the electric charge carried by one mole of electrons. These constants are intrinsic to the material and reaction under consideration, specifically the production of hydrogen and oxygen gases through the electrolysis of an aqueous electrolyte (Joswa Saputra et al. 2024; Kun et al. 2024; Mustari et al. 2021; Sari et al. 2022). The variable m, on the other hand, represents the mass of the substance produced at the electrode surface during the electrolysis process. In the context of hydrogen fuel cells, the formation of hydrogen and oxygen gases is crucial, as they are reactive products necessary to initiate the chemical reaction that produces electrical energy. The amount of gas produced directly affects the energy conversion efficiency and the system's output voltage.

By increasing the immersion depth, the fuel cell benefits from a larger reaction interface, which supports higher gas formation rates (hydrogen at the cathode and oxygen at the anode), resulting in better energy output. Furthermore, deeper immersion contributes to more stable and uniform ionic conduction pathways within the electrolyte, reducing resistance and enabling more efficient current flow. These findings underscore the importance of optimizing electrode exposure in the electrolyte when designing hydrogen fuel cells, particularly in systems where the surface area directly impacts performance.

$$m = \frac{Q.M}{n.F} (6)$$

The Q value, which refers to the total electrical charge passing through the system, plays a crucial role in determining the mass of the substance formed. The Q value is not constant, but instead calculated using Equations (7) and (8), where Q is defined as the product of current (I) and time (t). This relationship highlights the direct proportionality between the amount of charge supplied and the amount of gas produced, further reinforcing the importance of sufficient current flow and adequate charging duration in optimizing fuel cell performance.

$$Q = I \,.\, t (7)$$
$$I = \frac{V}{R} (8)$$

When only a small portion of the electrode is immersed in the electrolyte, the limited cross-sectional area in contact with the solution reduces the surface area available for electron transfer. This limitation hinders the electrolyte's ability to effectively facilitate ion exchange and release electrons at the electrode surface, resulting in lower overall electrochemical activity. As the immersed cross-sectional area increases, more surface area is available for redox reactions to occur, allowing for more efficient electron flow. According to Equation (7), the increased active area contributes to a higher electric current (I), which in turn results in a greater electric charge (Q) being transferred, as charge accumulation plays a crucial role in generating electrical energy as a larger immersed area yields a higher output voltage and enhanced overall fuel cell performance. In addition to electric current, Equation (7) highlights time (t) as another critical variable that influences the total electric charge (Q) produced during electrolysis. To evaluate the influence of this parameter, experiments were designed to include variations in charging duration, specifically at intervals of 30, 60, and 90 seconds. The goal was to determine whether extending the electrolysis time would result in a measurable increase in the output voltage of the hydrogen fuel cell. However, the results showed that increasing the charging duration did not result in a proportional increase in the output voltage, as might be theoretically expected under ideal conditions (Susilo, Gumono, and Setiawan 2023; Toruan et al. 2023). This difference is likely due to the unsealed configuration of the prototype fuel cell used in this study. Without a sealed system, the hydrogen and oxygen gas produced during electrolysis cannot be effectively contained and can quickly diffuse into the surrounding environment (Toshiba 2022; Wang et al. 2021). As a result, these gases are not retained at the electrode interface in sufficiently high concentrations or for a reasonably long duration to significantly enhance the redox reactions that produce electrical energy. Consequently, extended charging times fail to make a corresponding increase in output voltage. These findings highlight an essential consideration in the design of small-scale experimental fuel cells: although electric current and time are theoretically influential in determining the quantity of charge and mass of gases produced (as described by Faraday's Law), the storage and retention of these gases are equally important (Kun et al. 2024; Mustari et al. 2021). In systems that lack a gas-tight enclosure or pressure barrier, even well-planned electrical input may not yield increased energy. This highlights the need for closed or semi-pressurized fuel cell designs to ensure that the hydrogen and oxygen produced are retained and utilized more effectively in sustainable electricity production (Fontana 1986; Hassan et al. 2023; Kun et al. 2024; Singla et al. 2021).

The implications of this research provide a strong scientific basis for the potential use of fuel cells as a clean and sustainable energy source. These findings serve as a reference for the development of environmentally friendly alternative energy technologies, particularly in efforts to reduce dependence on fossil fuels and reduce carbon emissions. Furthermore, this research opens up opportunities for the application of hydrogen fuel cells in various sectors, such as transportation, household electricity supply, and portable devices. Academically, the results of this study can enrich the literature on the performance characteristics of fuel cells, while also providing a basis for further research focused on improving efficiency, voltage stability, and the development of more economical and durable electrode materials. The limitations of this research lie in the limited scope of experiments conducted on a laboratory scale, using relatively simple variables such as hydrogen concentration, temperature, and electrode surface area. This means that the research results do not fully reflect the real conditions in large-scale use. Additionally, the limited availability of electrode materials and cooling systems can also impact the accuracy and stability of the resulting voltage. Based on these limitations, it is recommended that further research develop fuel cell designs with more complex and realistic systems, involving more economical electrode material variations and long-term performance testing. Additional research is also recommended to integrate aspects of energy efficiency, hydrogen storage security, and production cost analysis, so that the results can be more applicable to support the development of environmentally friendly alternative energy sources.

4. CONCLUSION

Based on the experiments and data collected, the percentage of immersed electrode surface and the concentration of $NaHCO_3$ electrolyte are two significant factors that affect the voltage output in a hydrogen fuel cell system. The larger electrode surface area in direct contact with the electrolyte increases the rate of the electrochemical reaction. In contrast, the higher electrolyte conductivity facilitates more efficient charge transfer, both resulting in increased voltage. In contrast, the duration of hydrogen and oxygen production had a negligible impact on voltage under test conditions, indicating that those above geometric and chemical parameters primarily influence short-term performance. However, to fully evaluate the role of time-dependent factors (e.g., gas accumulation and Faraday efficiency), future studies should use closed reaction chambers. Fuel cell design: This modification will prevent gas diffusion into the environment through tiny gaps in the lid, allowing for the precise measurement of how prolonged operation affects energy conversion efficiency and storage capacity, as predicted by Faraday's Law. With a scalable electrode design and optimized NaHCO3 concentration, the gap between instantaneous voltage output and sustained fuel cell performance can be bridged efficiently.

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